



GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
Department of Computer Science and Engineering

Major Project Phase-I & II Review-1

Adaptive Combat Agent using Reinforcement Learning

Under the esteemed guidance of

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Contents

1. Introduction
2. Abstract and Domain
3. Objective and Problem statement
4. Literature Survey
5. Project requirements
6. Description of the project
7. Architecture and Design Process
8. Method/Algorithm/ Modules
9. Outcome (Execution of the Project)
10. References

Introduction

- Reinforcement Learning (RL) is a powerful deep learning technique that enables agents to acquire complex skills through trial-and-error interactions with dynamic environments. Unlike supervised methods, RL allows agents to learn strategies by maximizing rewards derived from performance outcomes, making it well-suited for adaptive decision-making tasks.
- This project applies RL to develop adaptive combat agents for both hand-to-hand and sword-fighting scenarios. Agents learn offensive and defensive skills without explicit supervision, guided by performance metrics such as winning rate, strike efficiency, survival time, and stamina management. Hand-to-hand agents adapt to strike-block-dodge situations, while sword agents master timing, positioning, and blade control.
- The system demonstrates the potential of RL to create realistic combat simulations and intelligent training environments, highlighting opportunities for future research in combat AI and autonomous agent design.

Abstract

- Reinforcement Learning is one of the powerful deep learning techniques for training autonomous agents to acquire complex skills via trial and error interactions with dynamic environments
- This project proposes the application of Reinforcement Learning for developing Adaptive combat behavioural agents, focusing on both hand to hand and sword fight styles.
- The proposed system enables agents to learn offensive and defensive skills without explicit supervision relying on reward signals that derive from combat performance metrics like winning rate, strike efficiency, survival time and stamina management during combat.
- Hand to hand combat agents will be able to adapt to strike-block-dodge scenarios accurately and sword agents will be able to handle timing, positioning and blade control.
- This project highlights the potential of realistic simulations, intelligent training systems and future research in combat AI.

Problem Statement

- Traditional combat agents' systems rely on predefined set of rules, scripted behaviours which limit adaptability and realism of the agents. They struggle to dynamically adjust strategies in response to varying opponents and combat scenarios. This lack of adaptability reduces both the training value of combat simulations and the immersion in interactive environments.
- Hand to Hand combat and sword fighting systems, in particular, should have rapid decision-making skills and tactical variations and fine-grained control over movements which are very difficult to hard-code or manually design and implement.
- The existing systems tend to repeat similar attack/defensive patterns, failing to exhibit situational awareness and adaptive defense. They fail to simulate the gradual learning progress of the human fighters
- This project aims to solve this problem by applying reinforcement learning to develop adaptive combat agents capable of learning offensive and defensive skills autonomously scaling across multiple difficulty levels and exhibiting human like combat strategies.

Objectives

- Develop adaptive combat agents using Reinforcement Learning for both hand-to-hand and sword-fighting scenarios.
- Design a reward mechanism based on combat metrics such as win rate, strike efficiency, stamina management, and survival time.
- Enable autonomous learning of offensive and defensive combat strategies without explicit supervision.
- Simulate realistic combat environments to train agents in strike–block–dodge interactions and weapon-based engagements.
- Demonstrate applications of combat AI in intelligent training systems, gaming, simulations, and future research.

Literature Survey

S.No.	Title of The Paper & Year	Authors	Methodology & Metrics	Datasets Used	Observed Shortcomings/ Gaps in The Paper
1.	Testbed for Action Fighting Game AI (2025)	Pakanan Aimherungilas, Vishnu Kotrajaras	Unity-based customizable 3D fighting testbed, AI via dynamic scripting & pattern-based models; adaptability & gameplay data	Custom Unity environment (no external dataset)	1. No jumping & projectile attacks 2. Limited AI flexibility 3. No multi-enemy support 4. Less detailed than commercial games
2.	Multi-Agent Reinforcement Learning for Creating Intelligent Agents in Social Networks-Oriented Role Playing Games (2025)	Francisco Martinez-Gil, Eduard Gil-Magraner	MARL for RPG NPCs; Comparison of DQN, PPO, QMIX; Specialized attack & defense learning strategy	Custom MUD-type game (text-based)	1. Text-based game (not realistic) 2. Simplified Combat 3. Subjective evaluation (Likert survey)
2.	Multi-Agent Reinforcement Learning in Games: Research and Applications (2025)	Haiyang Li, Ping Yang, Weidong Liu, Shaoqiang Yan, Xinyi Zhang, Donglin Zhu	Review of MARL in Games; Taxonomy of MARL (value function optimization, policy gradient learning, online search planning); Bio-inspired computing principles	N/A (review paper)	1. Theoretical Focus (no new results) 2. Broad Scope 3. Lack of Critical Analysis

Literature Survey

S.No.	Title of The Paper & Year	Authors	Methodology & Metrics	Datasets Used	Observed Shortcomings/ Gaps in The Paper
4.	Reinforcement Learning of Bipedal Walking Using a Simple Reference Motion (2024)	Naoya Itahashi, Hideaki Itoh, Hisao Fukumoto, Hiroshi Wakuya	RL with Simple Reference Motion; Bayesian Optimization for parameters; Inverse Kinematics for joint angles	KHR-3HV humanoid robot	1. Limited to specific robot 2. Simple reference motion 3. No obstacle avoidance
5.	Advancing DRL Agents in Commercial Fighting Games: Training, Integration, and Agent-Human Alignment (2024)	Chen Zhang, Qiang He, Yuan Zhou, Elvis S. Liu, Hong Wang, Jian Zhao, Yang Wang	Shukai DRL system, HELT, PPO, multi-style rewards; 22% ↑ efficiency; Elo score & retention eval	Naruto Mobile (400+ chars, 50 subset, 40M+ matches)	1. High computational cost 2. Weak generalization on unseen characters 3. Agents more cautious than humans 4. Limited to Naruto Mobile only 5. Needs expert priors for character selection
6.	A Study on the Agent in Fighting Games Based on Deep Reinforcement Learning (2022)	Hai Liang, Jiaqi Li	End-to-end frame system with DDQN, action space defined, reward shaping; 95% win rate	Street Fighter V	1. Limited to one game 2. Risk of overfitting repetitive actions 3. Unstable reward curves 4. Weak vs higher-level AI 5. No human player evaluation

Literature Survey

S.No.	Title of The Paper & Year	Authors	Methodology & Metrics	Datasets Used	Observed Shortcomings/ Gaps in The Paper
7.	A Deep Reinforcement Learning Blind AI in DareFightingICE (2022)	Thai Van Nguyen, Xincheng Dai, Ibrahim Khan, Ruck Thawonmas, Hai V. Pham	DRL agent using only sound input; Three audio encoders compared; Trained with PPO; evaluated via win rate & avg HP difference vs weakened MCTS AI	DareFightingICE platform	1. Tested only vs weakened AI, not full or human players 2. Limited sound cues for gameplay 3. Poor generalization to other sound designs
8.	Behavioral Cloning in the Fighting Game <i>The King of Fighters XV</i> (2022)	Alejandro J. Lugo, David Serrano, Francisco J. Rodriguez, David Camacho	Behavioral cloning via CNN; trained on expert gameplay state-action pairs; evaluated by win rate vs AI/humans & action prediction accuracy	Custom dataset (expert gameplay from <i>The King of Fighters XV</i>)	1. Limited to imitation, no new strategies 2. Performance capped by expert skill 3. Distributional shift issues in unseen states
9.	Creating Pro-Level AI for a Real-Time Fighting Game Using Deep Reinforcement Learning (2021)	Inseok Oh, Seungeun Rho, Sangbin Moon, Seongho Son, Hyoil Lee, Jinyun Chung	DRL agent for <i>Blade & Soul</i> ; self-play curriculum (aggressive/defensive/balanced); Data skipping for efficiency; Evaluated by win rate vs pro humans & human-likeness	Blade & Soul (commercial fighting game)	1. High computational cost 2. Game-specific, limited generalization 3. Ethical concerns about fair play & misuse

Literature Survey

S.No.	Title of The Paper & Year	Authors	Methodology & Metrics	Datasets Used	Observed Shortcomings/ Gaps in The Paper
10.	Developing Combat Behavior Through Reinforcement Learning (2020)	Jonathan A. Boron	RL for military combat simulation; aggregate-level constructive sim; evaluated on tactical principles (mass & economy of force) and hyperparameter impacts	Custom combat simulation	1. Simplified simulation lacking real-world complexity 2. Narrow scope, limited scenarios 3. No human-in-the-loop evaluation

Requirements

Software Requirements

- **Operating System:** Windows 10/11 (64-bit)
- **Game Engine:** Unreal Engine 5.6 (preferred for advanced physics, animations, and rendering)
- **Programming Languages & Frameworks:**
 - **C++** – Unreal core logic and custom RL integration
 - **Blueprints** – Rapid prototyping inside Unreal
 - **Python 3.11** – Reinforcement learning algorithms and training

Hardware Requirements

- **CPU:** Quad-core Intel i5 / AMD Ryzen 5 (3.0+ GHz)
- **Memory (RAM):** 16 GB or higher
- **GPU:** NVIDIA GTX 1060 / AMD equivalent
- **VRAM:** 4 GB (minimum)
- **Storage:** 256 GB SSD (for fast data access)